10.1 - Defining Convergent and Divergent Infinite Series

Learning Objective(s):
LIM-7.A: Determine whether a series converges or diverges.

Sequences

Consider the following *list* of numbers:

$$\{1, 4, 7, 10, 13, 16, \ldots\}$$

How would you suppose this list is generated?

This "list" is an example of a **sequence** where each number in the sequence is called a **term** of the sequence. We can denote a sequence in a variety of ways – three of the most popular are:

$$\{a_1, a_2, a_3, a_4, \ldots, a_n, \ldots\}, \quad \{a_n\}_{n=1}^{\infty}, \quad \text{and} \quad \{a_n\}$$

The subscript that appears with this sequence is called an **index** and it indicates the order of the terms in the sequence. A sequence will often start with an index of $n = 1$ but can sometimes begin with $n = 0$.

There are two ways to define a sequence as illustrated in the table below.

Defining a Sequence	
Recursive Definition (a.k.a. Implicit Definition)	**Explicit Definition**
Note: Each term in the sequence is three more than the preceding term. Therefore, $a_1 = 1$ $a_2 = a_1 + 3$ $a_3 = a_2 + 3$ $\vdots$ $a_{n+1} = a_n + 3$ Putting this all together, formally we should say: $a_1 = 1$ and $a_{n+1} = a_n + 3$ where $n = 1, 2, 3, \ldots$	**Note:** Each term in the sequence can be represented by ordered pairs using the convention: (order of the term, actual value of the term) Therefore, we can rename those terms with $(1,1), (2,4), (3,7), (4,10), \ldots$ and use a known relationship between the ordered pairs. Here, we have a linear relationship. $slope = \frac{4-1}{2-1} = \frac{7-4}{3-2} = \frac{10-7}{4-3} = \ldots = 3$ Using the *point-slope* formula with the first coordinate point $a_n - 1 = 3(n-1)$ or $a_n = 3n - 2$
Benefit: The relationship is typically easy to see.	**Benefit:** The explicit definition is extremely helpful in finding term deep into the sequence like a_{50}.
Drawback: The recursive definition is not very helpful if we want to find a term deep into the sequence like a_{50}.	**Drawback:** The relationship can be difficult to see if the sequence is not so "well-behaved", i.e., linear, or geometric.

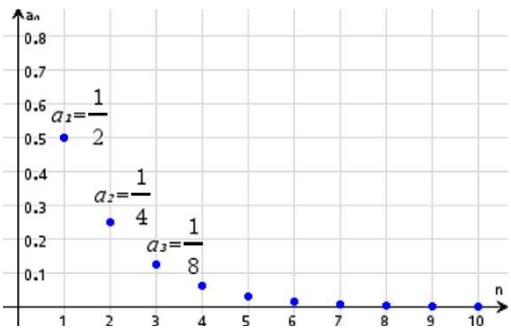
Example 1: Explicit Formulas

Use the explicit formula given for each sequence, $\{a_n\}_{n=1}^{\infty}$, to write out the first four terms of each sequence.

a. $a_n = \dfrac{1}{2^n}$ $\left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \ldots\right\}$

b. $a_n = \dfrac{(-1)^n n}{n^2+1}$ $\left\{-\frac{1}{2}, \frac{2}{5}, \frac{-3}{10}, \frac{4}{17}, \ldots\right\}$

Graphically, the two sequences in Example 1 would look like the following.



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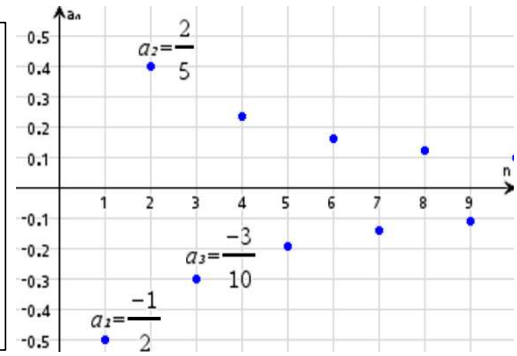

Example 2: Working with Sequences

For each of the following sequences, find

i.) the next two terms of the sequence.

ii.) a recursive definition that generates the sequence.

iii.) an explicit formula for the *n*th term of the sequence.

a. $\{a_n\} = \{-2, 5, 12, 19, \ldots\}$ 26, 33, ...

$a_1 = -2$ and $a_{n+1} = a_n + 7$

$a_n = 7n - 9$,

b. $\{b_n\} = \{3, 6, 12, 24, 48, \ldots\}$ 96, 192, . . .

$b_1 = 3$ and $b_{n+1} = 2b_n$

$b_n = 3(2)^{n-1}$

Limit of a Sequence

DEFINITION OF THE LIMIT OF A SEQUENCE

Let L be a real number. The **limit** of a sequence $\{a_n\}$ is L, written as

$$\lim_{n\to\infty} a_n = L$$

if the terms of the sequence approach a unique number, L, as n increases.

If the limit L of a sequence exists, then the sequence **converges** to L.
If the limit to a sequence does not exist, then the sequence **diverges**.

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THEOREM 10.1: LIMIT OF A SEQUENCE

Let L be a real number. Let f be a function of a real variable such that

$$\lim_{x\to\infty} f(x) = L$$

If $\{a_n\}$ is a sequence such that $f(n) = a_n$ for every positive integer n, then

$$\lim_{n\to\infty} a_n = L$$

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Example 3: Limit of a Sequence

For each of the following sequences, write out the first four terms. If you believe the sequence converges, make a conjecture about its limit. If the sequence appears to diverge, explain why. Assume each sequence has a beginning index of $n = 1$ and is defined infinitely.

a. $\{a_n\} = \left\{\frac{(-1)^n}{n^2+1}\right\} = \left\{-\frac{1}{2}, \frac{1}{5}, -\frac{1}{10}, \frac{1}{17}, \ldots\right\}$ $\lim_{n\to\infty} a_n = 0$ converges!	**b.** $\{b_n\} = \{\cos(n\pi)\} = \{-1, 1, -1, 1, -1, \ldots\}$ $\lim_{n\to\infty} a_n$ DNE diverges!
c. $\{c_n\} = \left\{\frac{4n^3}{n^3+1}\right\} \longrightarrow$ converges to 4 $= \left\{2, \frac{32}{9}, \frac{27}{7}, \frac{256}{65}, \ldots\right\}$ $\lim_{n\to\infty} \frac{4n^3}{n^3+1} = 4$	**d.** $d_{n+1} = -2d_n$, $d_1 = 1$ $\{1, -2, 4, -8, \ldots\}$ $\lim_{n\to\infty} a_n$ DNE diverges!

Example 4: A Bouncing Basketball

A basketball is tossed straight up in the air and reaches a high point before falling to the floor. Each time the ball bounces on the floor, it rebounds to 0.6 of its previous height. Let h_n be the high point after the nth bounce, with the initial height being 20 feet.

a. Find a recursive definition and an explicit formula for the sequence $\{h_n\}$.

$h_0 = 20$

$h_{n+1} = 0.6h_n$

$h_n = 20(0.6)^n$

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b. What is the high point after the 10th bounce?

$h_{10} = 20(0.6)^{10} = 0.121$ ft

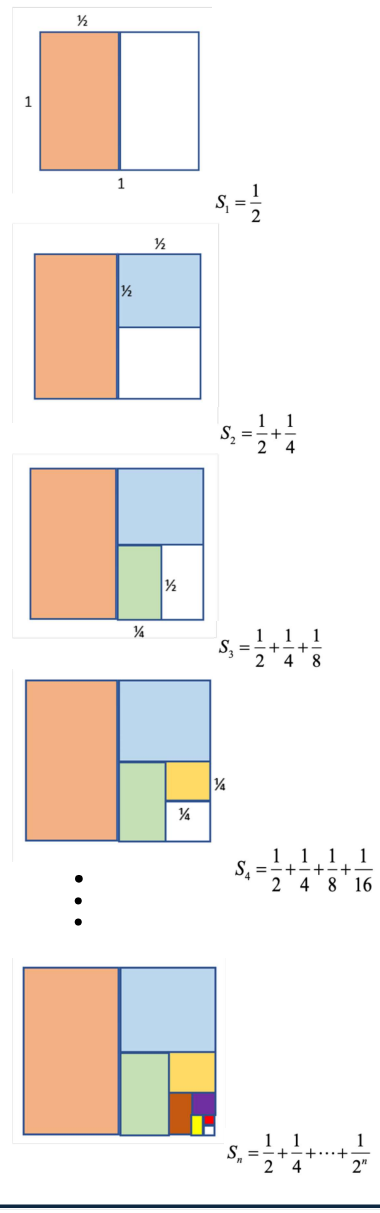
c. What conjecture can you make about the limit of the sequence $\{h_n\}$?

$\lim_{n\to\infty} h_n = 0$

Infinite Series and the Sequence of Partial Sums

An infinite series can be viewed as a sum of an infinite set of numbers and looks like $a_1 + a_2 + a_3 + \cdots + a_n + \cdots,$

where the terms of the series, $a_1, a_2, \ldots,$ are real numbers.



How is it possible, though, to add up an infinite set of numbers?

Consider a square with sides of length 1 that is subdivided as shown in the images to the right. We can let S_n be the area of the colored region in the nth figure of the progression.

The area of the colored region in the first figure is

$$S_1 = 1 \cdot \frac{1}{2} = \frac{1}{2}$$

$$\frac{1}{2} = \frac{2^1 - 1}{2^1}$$

The area of the colored region in the second figure is S_1 plus the area of the smaller blue square,

which is $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. Therefore,

$$S_2 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$\frac{3}{4} = \frac{2^2 - 1}{2^2}$$

The area of the colored region in the third figure is S_2 plus the area of the smaller green rectangle,

which is $\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$. Therefore,

$$S_3 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}$$

$$\frac{7}{8} = \frac{2^3 - 1}{2^3}.$$

If we continue in this manner, we discover that

$$S_n = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots \frac{1}{2^n} = \frac{2^n - 1}{2^n}$$

If we continue this process indefinitely, the area of the colored region, S_n approaches entire area of the square, which is 1. Therefore, it is possible to say

$$\lim_{n\to\infty} S_n = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1.$$

$$\lim_{n\to\infty} \frac{2^n - 1}{2^n} = \lim_{n\to\infty} \frac{2^n \ln 2}{2^n \ln 2} = 1$$

DEFINITION OF INFINITE SERIES

Given a sequence of numbers $\{a_1, a_2, a_3, \ldots\}$, the sum

$$a_1 + a_2 + a_3 + \cdots = \sum_{k=1}^{\infty} a_k$$

Is called an **infinite series**. Its **sequence of partial sums**, $\{S_n\}$, has the terms

$$S_1 = a_1$$
$$S_2 = a_1 + a_2$$
$$S_3 = a_1 + a_3 + a_3$$
$$\vdots$$
$$S_n = a_1 + a_2 + a_3 + \cdots + a_n = \sum_{k=1}^{n} a_k, \text{ for } n = 1, 2, 3, \ldots$$

If the sequence of partial sums, $\{S_n\}$, has a limit L, the infinite series **converges** to that limit, and we can write

$$\sum_{k=1}^{\infty} a_k = \lim_{n\to\infty} \sum_{k=1}^{n} a_k = \lim_{n\to\infty} S_n = L$$

If the sequence of partial sums diverges, the infinite series also **diverges**.

<u>Example 5</u>: Sequence of Partial Sums

Consider the infinite series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$. $= 1$

$S_4 = \frac{3}{4} + \frac{1}{20} = \frac{4}{5}$

a. Find the first four terms of the sequence of partial sums.

$S_1 = a_1 = \frac{1}{2}$ $\quad S_2 = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}$ $\quad S_3 = \frac{2}{3} + \frac{1}{12} = \frac{3}{4}$

b. Find an expression for S_n and make a conjecture about the value of the series.

$S_n = \frac{n}{n+1}$ $\quad \lim_{n\to\infty} \frac{n}{n+1} = 1$

Scan the QR Code above to watch a video covering Example 5

<u>Example 6</u>: A Sequence Versus a Series

Consider the following sequence defined by $a_n = \frac{n}{2n+1}$ and its corresponding series defined by

$S_n = \sum_{k=1}^{n} \frac{k}{2k+1}$.

Find each of the following limits.

a. $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2}$

b. $\lim_{n\to\infty} S_n = \sum_{n=1}^{\infty} \frac{n}{2n+1} = \infty$ (diverges!)

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10.2 - Working with Geometric Series

Learning Objective(s):
LIM-7.A: Determine whether a series converges or diverges.

1. A geometric series does **not** have to begin with an index of $n = 0$.

2. A more intuitive way to compute the sum of a convergent geometric series is to think of the sum as

$$S = \frac{\text{first term of the series}}{1 - \text{the common ratio}}$$

Geometric Series

Specific Example: $3 + 3\cdot 2 + 3\cdot 2^2 + 3\cdot 2^3 + \cdots$

General Form: $\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + \cdots + ar^n + \cdots, \quad a \neq 0$

THEOREM 10.2: CONVERGENCE OF A GEOMETRIC SERIES

1. A geometric series with a ratio r diverges if $|r| \geq 1$.

2. If $0 < |r| < 1$, then the series converges to the sum:

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}, \quad 0 < |r| < 1.$$

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Example 1: Identifying Geometric Series
State whether or not each of the following series are geometric.

a.) $\sum_{n=1}^{10} \left(\frac{1}{2}\right)^n =$ ✓

$a = \frac{1}{2}$ $r = \frac{1}{2}$

b.) $\sum_{n=1}^{20} \frac{1}{n}$ ✗

Harmonic

c.) $\sum_{n=0}^{20} (2n+1)$ ✗

Example 2: **Convergent and Divergent Geometric Series**
Evaluate each of the following geometric series or state the series diverges.

a.) $\sum_{n=0}^{\infty} (1.1)^n$ $r = 1.1 > 1$

diverges!

$\lim_{n\to\infty} (1.1)^n = \infty \neq 0$

b.) $\sum_{n=0}^{\infty} e^{-n} = \sum_{n=0} \left(\frac{1}{e}\right)^n$

$\frac{1}{e} < 1$ converge!

$= \frac{1}{1 - \frac{1}{e}} = \frac{e}{e-1}$

c.) $\sum_{n=1}^{\infty} 3(-0.75)^n$ $r = |-0.75| < 1$ converges!

$= \frac{-\frac{9}{4}}{1 - \left(-\frac{3}{4}\right)} = -\frac{9}{7}$

c.) $\sum_{n=0}^{\infty} \frac{3}{2^n} = \sum_{n=0}^{\infty} 3\left(\frac{1}{2}\right)^n$

$\left|\frac{1}{2}\right| < 1$ Converges

$= \frac{3}{1 - \frac{1}{2}} = 6$

d.) $\sum_{n=0}^{\infty} \left(\frac{3}{2}\right)^n$

Diverges!

e.) $\sum_{n=0}^{\infty} \left(\frac{e}{\pi}\right)^n = \frac{1}{1 - \frac{e}{\pi}}$

$= \frac{\pi}{\pi - e}$

Example 3: Decimal Expansion as a Geometric Series

Write $1.0\overline{25}$ as a geometric series and express its value as a fraction.

$$1.0\overline{25} = 1 + \left(\frac{25}{1000} + \frac{25}{100000} + \frac{25}{10000000} + \cdots\right)$$

$$1.025252525\ldots = 1 + \frac{25}{1000} \cdot \frac{1000}{1000} = 1 + \frac{25}{990} = \frac{1024}{990} = \frac{512}{495}$$

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Telescoping Series

Although telescoping series are not a part of the AP Calculus BC curriculum and will not be tested on the AP exam, they are an important member of the series family. We will take a brief look at this type of series.

Telescoping series have the form

$$(b_1 - b_2) + (b_2 - b_3) + (b_3 - b_4) + (b_4 - b_5) + \cdots \quad \text{therefore } S_n = b_1 - b_{n+1}$$

A telescoping series converges if and only if $b_n \to$ a finite number as $n \to \infty$. In which case

$$S = b_1 - \lim_{n\to\infty} b_{n+1}$$

1. Many telescoping series contain several terms b_1, b_2, b_3,... that do not cancel and become part of the sum.
2. Partial fraction decomposition is often used to rewrite the series.

Example 4: Telescoping Series

Find the sum of the following series.

a.) $\displaystyle\sum_{n=1}^{\infty}\left(\frac{1}{2^n} - \frac{1}{2^{n+1}}\right) = \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{8}\right) + \left(\frac{1}{8} - \frac{1}{16}\right) + \cdots$

$= \frac{1}{2}$

b.) $\displaystyle\sum_{n=1}^{\infty}\left(\frac{1}{n(n+1)}\right) = \sum_{n=1}^{\infty}\left(\frac{1}{n} - \frac{1}{n+1}\right) = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots$

$= 1$

$$\frac{1}{n(n+1)} = \frac{A}{n} + \frac{B}{n+1}$$

$$1 = A(n+1) + Bn$$

$n=0$: $A = 1$

$n=-1$: $1 = -B \Rightarrow B = -1$

Example 5: Writing a Series in Telescoping Form

Write the following series in telescoping form and find the sum of

$$\sum_{n=1}^{\infty}\frac{2}{n^2+4n+3}. = \sum_{n=1}^{\infty}\left(\frac{1}{n+1} - \frac{1}{n+3}\right)$$

$$= \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \left(\frac{1}{4} - \frac{1}{6}\right) + \left(\frac{1}{5} - \frac{1}{7}\right)$$

$$= \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

$$\frac{2}{(n+1)(n+3)} = \frac{A}{n+1} + \frac{B}{n+3}$$

$$2 = A(n+3) + B(n+1)$$

$n=-1$: $2 = 2A \to A = 1$

$n=-3$: $2 = -2B \Rightarrow B = -1$

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Scan the QR Code above to watch a video covering Example 5

10.3 - The nth Term Test for Divergence

Learning Objective(s):
LIM-7.A: Determine whether a series converges or diverges.

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*n*th-Term Test for Divergence

This topic begins with a couple theorems.

THEOREM 10.3A: LIMIT OF THE *n*TH TERM OF A CONVERGENT SERIES

$$\text{If } \sum_{n=1}^{\infty} a_n \text{ converges, then } \lim_{n\to\infty} a_n = 0$$

It seems rather intuitive that a series cannot possibly add to a finite sum unless the terms themselves approached 0. What is important from the above theorem is that its contrapositive is true as well.

THEOREM 10.3B: *n*TH TERM TEST FOR DIVERGENCE

$$\text{If } \lim_{n\to\infty} a_n \neq 0, \text{ then } \sum_{n=1}^{\infty} a_n \text{ diverges.}$$

Example 1: Using the nth-Term Test for Divergence
Determine the convergence or divergence of each series.

a.) $\sum_{n=0}^{\infty} \frac{n}{n+1}$ diverges by n^{th} term test

$\lim_{n\to\infty} \frac{n}{n+1} = 1 \neq 0$

b.) $\sum_{n=1}^{\infty} \frac{1+3^n}{2^n}$ diverges

$\lim_{n\to\infty} \frac{1+3^n}{2^n} = \lim_{n\to\infty} \frac{3^n \cdot \ln 3}{2^n \cdot \ln 2} = \infty$

c.) $\sum_{n=1}^{\infty} \frac{1}{n}$???

$\lim_{n\to\infty} \frac{1}{n} = 0$

d.) $\sum_{n=1}^{\infty} \frac{n^3}{n!}$???

$\lim_{n\to\infty} \frac{n^3}{n!} = 0$

1. It is very important to note that if $\lim_{n\to\infty} a_n = 0$ the series does not necessarily converge. This just means the **nth Term Test** is inconclusive.
2. You will start to see more factorials in your series expressions. They can be somewhat tricky to deal with as they "do not play nicely" with the calculus that we know. Thinking of the logical impact they have on the series expression is a good strategy.